# Quantitative Workflows

## Getting The Most From Futures Data
## FINM 37000 - Futures and Related Derivatives

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# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

1. Trading idea proposed or issue raised
2. Plan Analysis
3. Code to perform analysis
4. Analyze and evaluate
5. Build analysis into production level deliverable
6. Validate analysis in production test environment
7. Go live

# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

1. Trading idea proposed or issue raised

    - Gather additional information to ensure full understanding
    - Document proposal or issue for team visibility
    - Get feedback early and often to make sure all stakeholders are aligned on the big picture goals.
    - Some stakeholders may resist defining final goal until ideas have been validated, but the earlier in the process you can understand the goal, the better you can prepare and decide on tradeoffs along the way.

# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

2. Plan Analysis
    - Define quantifiable goal/result/measure of success
    - Data source(s)
    - Data preparation steps
        - Usual and custom transformations
        - Additional measurement processes
        - Fully specify preparation needed for inputs and outputs
    - Describe analysis connecting inputs and outputs
        - Highlight important aspects of the analysis that can be used for validation

# Evaluation

## How will you know your model is good

- Define your target
    - Simple cases: returns or price change
    - More realistic: measure with your execution strategy in mind
    - Distilling information to inform traders: model accuracy
- Estimate how well you hit your target
- Compare against baseline(s)

# P&L

## The goal is not the goal

- Every trading strategy is looking to generate positive P&L
- Low or negative P&L, real or simulated, does not tell you how to improve your model.
- High level stake holders will evaluate P&L.
- For simple trades, returns or price changes may approximate your P&L and can be the target of the analysis.
- More detailed metrics increase your ability to understand and improve model performance.

# Validation/Monitoring

## How will you know your model is working as expected

- Software tests pass
- Plot the data
- Fake data produces expected results
- Parameter estimates make sense
- Sample probabilities are at least approximately calibrated
- The point is not to accept or reject modeling assumptions, but measurements and visualizations of those assumptions help guide validation.

"All model are wrong, but some are useful." - George Box

# Evaluation as Validation/Monitoring

## Continue checking your model

- Evaluation procedure should be continued in production
  - Becomes another piece of validation/monitoring
- Time-series cross-validation
  - Fit model on rolling or expanding window of time
  - Evaluate performance on rolling out-of-sample data
  - Hyndman's Forecasting: Principles and Practice
  - Leave-future-out cross-validation
  - Python support sklearn.model_selection.TimeSeriesSplit, sktime.evaluate

# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

3. Code to perform analysis
    - Find or develop software to perform steps in the analysis plan
        - Don't build something yourself that already has a reliable library
        - Don't limit yourself by what is available in libraries
    - Test against basic cases to validate your implementation
        - Integration test: end-to-end analysis pipeline of plan from step 2
        - Unit test: isolated tests of individual components in the process
    - If there are many components to build, consider validating a simplified proof of concept as a first step.

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4. Analyze and evaluate

- Run fake data that you control through your pipeline.
    - Confirm analysis and evaluation work correctly.
- Run real data through your pipeline
    - Evaluate performance
        - This is the crux of academic experience, but the rest of the steps are key for transition to real world.
- Share results with stakeholders from step 1 to determine next steps

# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

5. Build analysis into production level deliverable
   - This phase is highly variable depending on deliverable
   - What can be reused from the research phase?
     - Some tradeoffs between speed of research and this step
     - Example 1: a microservice information or visualization dashboard for trading.
       - Good planning and design during the research phase can be reused for delivery.
     - Example 2: integration of analysis into high-frequency trading software
       - Often requires additional steps to adapt research pipeline to trading pipeline

# The Research to Production Pipeline

## An Linear Example of a Nonlinear Process

6. Validate analysis in production test environment
    - Ideally there is some test environment to validate new tools work as expected
    - Confirm that expectations and evaluations from research continue to hold
7. Go live
    - Continue logging and monitoring in production to identify when models break down.

# The Research to Production Pipeline

## Managing a Nonlinear Process

- Any of the above steps may fail requiring re-planning the process
- Shared documented proposal allows all stakeholders to understand progress and setbacks
    - Ideally, there are different levels of visibility on progress
    - Version control platforms are ideal for coordinating with developers
    - CEO/Head Trader do not care about your software revisions, so better to have a higher level, less noisy documentation of progress.

# Other Sources On Workflow

## Many fields face similar issues

- Most resources on workflow are not specific to trading or quantitative research, but we can learn from their experience.
- Software development: lots of buzzword overhead (Agile, Scrum, Kanban). Adapt to your team and understand what works for you. Example overview from Atlassian
- Machine learning:
  - Examples: ML Workflows, Geron's ML Book
- Bayesian statistics: Bayesian Workflow

# Process Details

## From Bayesian Workflow

- Potential workflow
- Focused on steps
    2. Plan analysis
    4. Analyze and evaluate
- Strategies for progress